EUROPEAN COMMISSION
DIRECTORATE-GENERAL
TAXATION AND CUSTOMS UNION
Indirect Taxation and Tax Administration
CBAM, Energy and Green Taxation

Brussels, 14 August 2026

GUIDANCE DOCUMENT 5E: SECTOR-SPECIFIC GUIDANCE DOCUMENT ON ALUMINIUM

Disclaimer: This guidance document is not legally binding and it is of an explanatory nature. Legal provisions of Union law take precedence over the contents of this document and should always be consulted. The authentic texts of the EU legal instruments are those published in the Official Journal of the European Union. This guidance was prepared by Commission services and does not represent an official position of the Commission. Only the Court of Justice of the European Union is competent to authoritatively interpret Union law.

VERSION HISTORY

Date	Version notes
14.08.2026	First Publication of this document for the definitive period.

CONTENTS

1 PREFACE 4

1.1 Structure and content of the guidance documents 4

1.2 Reference regulations and legal acts 5

2 CBAM GOODS AND PRODUCTION ROUTES 7

2.1 Foreword to sector-specific sections 7

2.2 Aluminium sector 8

2.2.1 Unit of production and embedded emissions 8

2.2.2 Definition and explanation of sector goods covered 9

2.2.3 Definition and explanation of relevant production processes and routes 11

3 ALUMINIUM SECTOR-SPECIFIC REQUIREMENTS 18

3.1 Sector-specific requirements for monitoring and reporting 18

3.1.1 Emissions monitoring 18

3.1.2 Additional rules 19

3.1.3 Additional reporting requirements 23

4 WORKED EXAMPLE FOR THE ALUMINIUM SECTOR 23

ANNEX A LIST OF DEFINITIONS 29

ANNEX B LIST OF ABBREVIATIONS 38

1 PREFACE

1.1 Structure and content of the guidance documents

This series of documents has been written to support stakeholders by explaining the requirements of the CBAM Regulation in a non-legislative language. This series of guidance documents should benefit a variety of stakeholders. A specific focus has been put on clarifying the **requirements for operators of installations producing CBAM goods outside of the EU for the definitive period**.

- **Guidance 1** provides an overview of the main changes to the CBAM methodology between the definitive period and the concluded transitional period. It also introduces the CBAM and an overview of the compliance cycle, roles and responsibilities, as well as milestones and deadlines for operators of installations outside the EU during the definitive period and explains the general exemptions from the CBAM.

- **Guidance 2** provides a quick guidance for the intended reader of this document, the operator of an installation producing CBAM goods. It provides a roadmap of the most important concepts of CBAM emissions monitoring and where to find more information in this series of documents.

- **Guidance 3** sets out the monitoring and reporting obligations and recommendations which are potentially applicable to any affected producer of CBAM goods.

- **Guidance 4** provides an overview of how the adjustment for free allocation is to be determined and what information is needed.

- **Guidance documents 5a, 5b, 5c, 5d, 5e and 5f** present an overview per sector of the production processes and value chains for goods that are included in the scope of the CBAM, as well as sector-specific monitoring and reporting considerations for each CBAM good where this is relevant. These sections are supplemented by examples for each sector.

The series of guidance documents is accompanied by an electronic template for information that should be used by installation operators to communicate information to the authorised CBAM declarants and CBAM Registry.

Presentation of numbers in EU documents

To align with EU legal documents, this guidance document uses the following convention when presenting numbers.

The decimal separator used to separate the integral part of a number from its fractional part is a comma, e.g.: 0,890

Thousands, and powers of 10^{3n} thereafter, are separated by a space, e.g.:

- fifteen thousand is written as 15 000
- fifteen million is written as 15 000 000

1.2 Reference regulations and legal acts

Several regulations and legal acts are referred to throughout this series of guidance documents. For ease of reading and to avoid ambiguity, simplified references are used when citing these acts.

- **CBAM Regulation**: Regulation (EU) 2023/956 and Regulation (EU) 2025/2083 amending Regulation (EU) 2023/956. Notably, where article numbers are given in this document without further specification, they always refer to the CBAM Regulation.

  It can be accessed at the following link (consolidated version):
  http://data.europa.eu/eli/reg/2023/956/2025-10-20

- **Methodology Act:** Commission Implementing Regulation (EU) 2025/2547, in legal texts referred to as "Implementing act pursuant to Article 7(7) of the CBAM Regulation".

  It can be accessed at the following link:
  http://data.europa.eu/eli/reg_impl/2025/2547/oj

- **Default Values Act**: Commission Implementing Regulation (EU) 2025/2621, in legal texts referred to as "Implementing act pursuant to Article 7(7) of the CBAM Regulation", also referred to as "default values determined in accordance with Annex IV to the CBAM Regulation".

  It can be accessed at the following link: **http://data.europa.eu/eli/reg_impl/2025/2621/oj**

- **Verification Principles Act**: Commission Implementing Regulation (EU) 2025/2546, in legal texts referred to as "Implementing act pursuant to Article 8(3) of the CBAM Regulation".

  It can be accessed at the following link:
  http://data.europa.eu/eli/reg_impl/2025/2546/oj

- **Accreditation & Verification Requirements Act**: Commission Delegated Regulation (EU) 2025/2551, in legal texts referred to as "Delegated act pursuant to Article 18(3) of the CBAM Regulation".

  It can be accessed at the following link: **http://data.europa.eu/eli/reg_del/2025/2551/oj**

- **Free Allocation Adjustment Act**: Commission Implementing Regulation (EU) 2025/2620, in legal texts referred to as "Implementing act pursuant to Article 31(2) of the CBAM Regulation".

  It can be accessed at the following link:
  http://data.europa.eu/eli/reg_impl/2025/2620/oj

For acronyms and definitions used in this document, please see **Annex B** and **Annex A**.

A series of icons are used throughout to help guide the reader:

Icon	Description of use
	Points to information of particular importance for operators of installations producing CBAM goods.
Simplified!	Highlights simplified approaches of the general requirements of the CBAM.
	Used where recommended improvements are presented.
	Used where other documents, templates or electronic tools are available from other sources.
	Points to examples given for the topics discussed in the surrounding text.

2 CBAM GOODS AND PRODUCTION ROUTES

This section provides guidance on industry-sector-specific rules that apply for the definitive period (from 2026 onwards), for aluminium sector. It deals with the specification of products covered by the CBAM and the relevant production routes.

2.1 Foreword to sector-specific sections

The following sections provide an overview of the different production routes for the goods under aluminium sector in CBAM and provide sector specific guidance. Where goods to which the same functional unit applies are produced in one installation via different production routes, the embedded emissions shall be determined in accordance with the averaging rules applicable in the definitive period.

Additional information on the production processes of the goods can also be found in the BREF[1] documents for best available techniques (BAT).

Diagrams used in the following sections.

For the system boundary graphics presented in the sections below, the **following conventions** are applied:

- **Production processes** (for which monitoring of the direct emissions would take place) are shown as **rectangles with white background**.
- **Optional** processes (e.g. CCS/CCU) are shown in **blue** boxes. In particular, CCS/CCU would not be taken into account for developing default values, but where you (as an operator) use them, the related emissions or emission savings should be taken into account for determining actual embedded emissions in line with the CBAM Regulation and the Methodology Act.
- **Materials** which are considered to have no embedded emissions are shown in **pink** boxes, materials with embedded emissions (precursor materials and final products, i.e. goods under the CBAM) in **green** boxes. Simple goods are shown in **normal** font, complex goods in **bold** font. Materials which are **optional** are shown in boxes with a **blue dashed** outline and must be included where actually used at the installation.
- Input materials are presented without trying to be complete. This means that the focus is on materials which are relevant for demonstrating the differences between different production routes. As a consequence, less important input materials and in particular fuels are usually omitted in order to keep graphs simple.
- **Electricity** as input is shown only in cases where it is the main "precursor" of the process (i.e. in particular for electric arc furnaces and electrolysis processes). Electricity is shown in **light grey** boxes.

[1] BAT Reference document (BREF), BAT being "Best Available Techniques" as defined by the IED (Industrial Emissions Directive). Relevant BREF documents are those for: the production of cement; for iron and steel production; large volume inorganic chemicals (which includes fertilisers); for Chlor-alkali; and for non-ferrous metals (which includes both aluminium and ferro-alloys). All BREFs can be found at the European IPPC Bureau, under https://eippcb.jrc.ec.europa.eu/reference.

2.2 Aluminium sector

The textbox below signposts the sector-specific sections in the Methodology Act, relevant for the CBAM definitive period.

Methodology Act references:

- **Annex I** ("Mapping of CN codes to aggregated goods categories") to the Methodology Act.
- **Annex I,** points 3.17 ("Unwrought aluminium") and 3.18 ("Aluminium products") which define the functional unit, system boundaries and possible precursors for aluminium sector goods.

2.2.1 Unit of production and embedded emissions

The quantity of declared aluminium goods imported into the EU must be expressed in metric tonnes. As an operator, you must record the quantity of CBAM good(s) produced by the installation or production process, for the purposes of reporting. In the definitive period, the reporting period corresponds to the calendar year of production, in accordance with Article 1(5) of the Methodology Act.

Industrial sector	**Aluminium**
Production unit of goods	Tonnes (metric), reported separately for each type of sector goods, by installation or production process in the country of origin.
Associated activities	Producing unwrought aluminium from alumina, or secondary raw materials (aluminium scrap), by metallurgical, chemical or electrolytic means; manufacture of semi-processed and finished aluminium products.
Relevant greenhouse gases	Carbon dioxide (CO_2) and perfluorocarbons (CF_4 and C_2F_6)
Direct Emissions	Tonnes (metric) of CO_2e
Indirect Emissions	Quantity of electricity consumed (MWh), source and emission factor used to calculate the indirect emissions in tonnes (metric) of CO_2 or CO_2e. During the transitional period, indirect emissions resulting from electricity consumption were monitored and reported separately. In the definitive period, for aluminium CBAM goods which are listed in Annex II ("Goods for which only direct emissions are taken into account") to the CBAM Regulation, only direct emissions are taken into account in their embedded emissions under the CBAM; indirect emissions of electricity are not included.
Unit for embedded emissions	Tonnes CO_2e emissions per tonne of goods, reported separately for each type of good, by installation in the country of origin.

During the transitional period, the aluminium sector reported both direct emissions and indirect emissions, with indirect emissions monitored and reported separately. For the definitive period, aluminium CBAM goods are listed in Annex II to the CBAM Regulation ("List of goods for which only direct emissions are to be taken into account, pursuant to Article 7(1)"). For those goods, only direct emissions are taken into account when determining embedded emissions for CBAM purposes; indirect emissions of electricity are not included, in line with Article 7(1) and Annex II of the CBAM Regulation and the cross-sectoral rules on system boundaries and precursors in the Methodology Act. Emissions must be reported in metric tonnes CO_2 equivalent (t CO_2e) emissions per tonne of output. This figure must be calculated for the specific installation or production process in your country of origin.

The following sections set out how the system boundaries of aluminium sector goods should be defined and identify elements of the production process that must be included for the purposes of monitoring and reporting.

2.2.2 Definition and explanation of sector goods covered

The following Table 2-1 lists the relevant goods in scope for the CBAM definitive period in the aluminium industry sector. The CN codes in the central column are the starting point for which 'production processes' are to be defined are to be defined for the purpose of monitoring in line with Annex I to the CBAM Regulation and Annex I ("Mapping of CN codes to aggregated goods categories") to the Methodology Act. Where goods with different CN codes to which the same functional unit applies are produced using different production routes within an installation, a single production process shall be used encompassing all production routes.

Table 2-1: CBAM goods in the aluminium sector[2]

Aggregated goods category	Product CN Code[3]	Description
Unwrought aluminium	7601	Unwrought aluminium

[2] Source: Annex I to the CBAM Regulation; Annex I ("Mapping of CN codes to aggregated goods categories") and points 3.17 ("Unwrought aluminium") and 3.18 ("Aluminium products") of Annex I to the Methodology Act.

[3] Where the CN codes indicated in this table consist of 4, 5 or 6 digits, all 8-digit CN codes sharing the same initial digits shall be considered included, unless otherwise specified.

Aggregated goods category	Product CN Code[3]	Description
Aluminium products	7603 – 7608, 7609 00 00, 7610, 7611 00 00, 7612, 7613 00 00, 7614, 7616	7603 – Aluminium powders and flakes
		7604 – Aluminium bars, rods and profiles
		7605 – Aluminium wire
		7606 – Aluminium plates, sheets and strip, of a thickness exceeding 0,2 mm
		7607 – Aluminium foil (whether or not printed or backed with paper, paper-board, plastics or similar backing materials) of a thickness (excluding any backing) not exceeding 0,2 mm
		7608 – Aluminium tubes and pipes
		7609 00 00 – Aluminium tube or pipe fittings (for example, couplings, elbows, sleeves)
		7610 – Aluminium structures (excluding prefabricated buildings of heading 9406) and parts of structures (for example, bridges and bridge-sections, towers, lattice masts, roofs, roofing frameworks, doors and windows and their frames and thresholds for doors, balustrades, pillars and columns); aluminium plates, rods, profiles, tubes and the like, prepared for use in structures
		7611 00 00 – Aluminium reservoirs, tanks, vats and similar containers, for any material (other than compressed or liquefied gas), of a capacity exceeding 300 litres, whether or not lined or heat-insulated, but not fitted with mechanical or thermal equipment
		7612 – Aluminium casks, drums, cans, boxes and similar containers (including rigid or collapsible tubular containers), for any material (other than compressed or liquefied gas), of a capacity not exceeding 300 litres, whether or not lined or heat-insulated, but not fitted with mechanical or thermal equipment
		7613 00 00 – Aluminium containers for compressed or liquefied gas
		7614 – Stranded wire, cables, plaited bands and the like, of aluminium, not electrically insulated
		7616 – Other articles of aluminium

The aggregated goods categories listed in Table 2-1 include both finished aluminium products and a precursor ‘unwrought aluminium’ that is itself used in the production of aluminium goods. The system boundaries specified in the Methodology Act do not mention explicitly which precursors are relevant for the production process. Therefore, Table 2-2 should help to identify the possible precursors by aggregated goods category and production route.

Table 2-2: Aggregated goods categories, their production routes and possible precursors

Aggregated Goods Category ***Production route***	**Precursors**
Unwrought aluminium *Primary aluminium* *Secondary aluminium*	None for primary aluminium For secondary aluminium – unwrought aluminium from other sources, if used in the process[4]
Aluminium products	Unwrought aluminium (differentiated between primary and secondary aluminium, if known), other aluminium products (if used in the production process).

Unwrought aluminium is produced by several production routes ('primary aluminium' for electrolytic smelting, 'secondary aluminium' for melting/recycling of scrap) as metal ingots, blocks, billets, slabs or similar. It is defined as a 'simple good', as the raw materials (carbon anodes and alumina for primary aluminium, scrap for secondary aluminium) used in its manufacture are themselves not treated as precursors (i.e. their upstream embedded emissions are not included).

The aluminium goods listed above include most types of aluminium product manufactured[5]. Aluminium products are defined as complex goods as they include the embedded emissions from the precursor unwrought aluminium.

The production of aluminium sector goods is by a number of different process routes, outlined below.

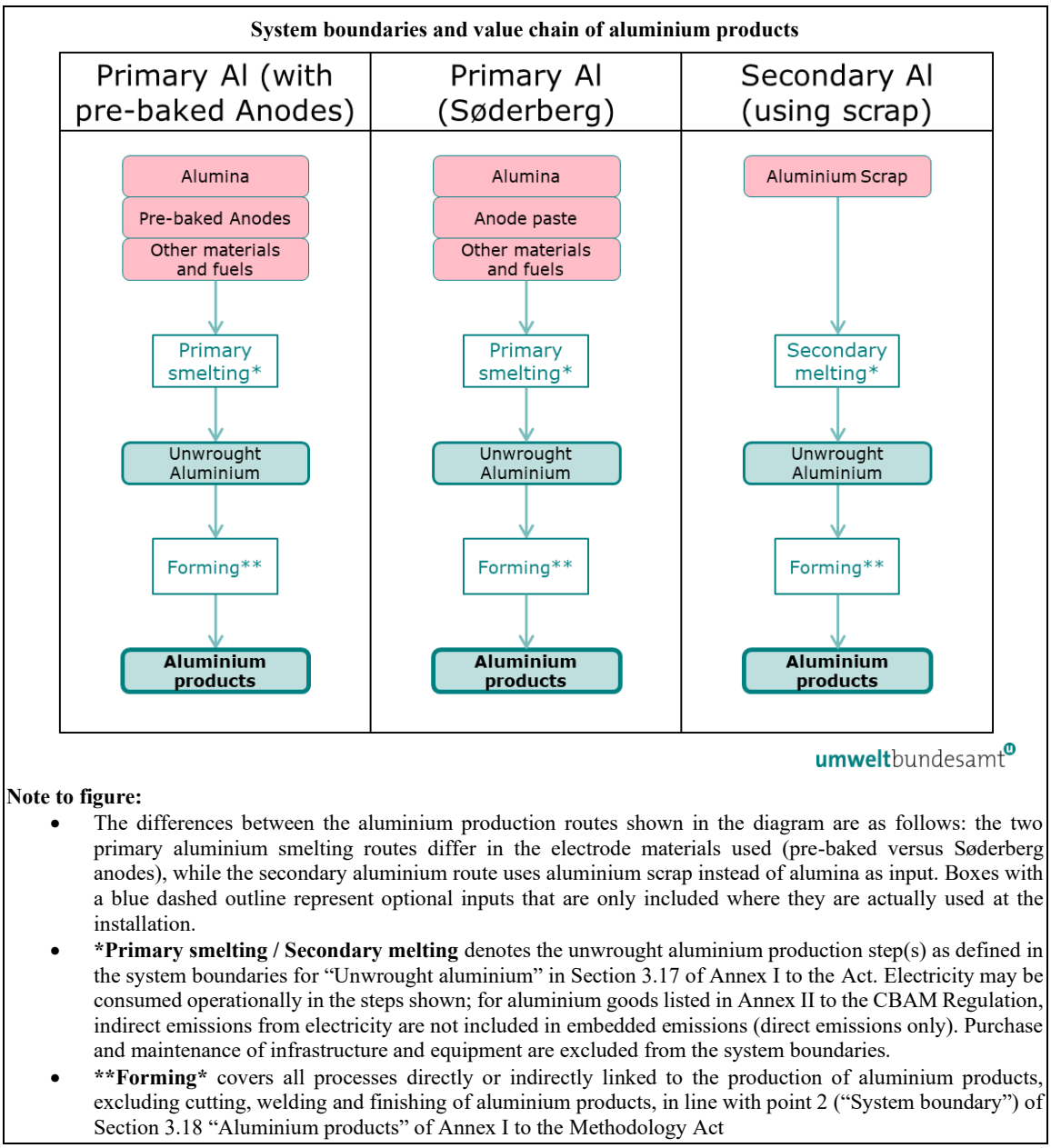
2.2.3 Definition and explanation of relevant production processes and routes

The system boundaries of the precursor unwrought aluminium and of aluminium products are distinct and may, under certain conditions, be added together to include all processes directly or indirectly linked to the production processes for these goods, including input activities to, and output activities from the process. In line with Annex I, point 3.1 (cross-sectoral rules) of the Methodology Act, the purchase and maintenance of infrastructure and equipment are excluded from the system boundaries.

[4] Note that any alloying elements contained in unwrought aluminium, that are not defined as CBAM precursor goods, must be excluded from the determination of mass and embedded emissions of the aluminium good.

[5] Excludes CN 7615 (certain household articles) and CN 7602 00 (aluminium waste and scrap).

Figure 2-1: System boundaries and value chain for unwrought aluminium and aluminium products



Note to figure:

- The differences between the aluminium production routes shown in the diagram are as follows: the two primary aluminium smelting routes differ in the electrode materials used (pre-baked versus Søderberg anodes), while the secondary aluminium route uses aluminium scrap instead of alumina as input. Boxes with a blue dashed outline represent optional inputs that are only included where they are actually used at the installation.
- ***Primary smelting / Secondary melting** denotes the unwrought aluminium production step(s) as defined in the system boundaries for "Unwrought aluminium" in Section 3.17 of Annex I to the Act. Electricity may be consumed operationally in the steps shown; for aluminium goods listed in Annex II to the CBAM Regulation, indirect emissions from electricity are not included in embedded emissions (direct emissions only). Purchase and maintenance of infrastructure and equipment are excluded from the system boundaries.
- ****Forming*** covers all processes directly or indirectly linked to the production of aluminium products, excluding cutting, welding and finishing of aluminium products, in line with point 2 ("System boundary") of Section 3.18 "Aluminium products" of Annex I to the Methodology Act

The differences between the aluminium production routes shown in the diagram are as follows: the two primary aluminium smelting routes differ in the electrode materials used (pre-baked anodes versus anode paste (Søderberg)), while the secondary aluminium route uses aluminium scrap instead of alumina as input.

The relevant emissions that must be monitored for the aluminium sector are detailed in section 3.1.1 of this document.

2.2.3.1 Unwrought aluminium – Primary (electrolytic) smelting production route

Primary aluminium is produced by the electrolysis of alumina[6] in electrolytic cells. During electrolysis, aluminium is reduced and oxygen from the alumina is liberated and combines with the carbon anode to form carbon dioxide and carbon monoxide – the carbon anodes in the primary aluminium process are therefore continuously consumed during the process.

Primary aluminium cell systems vary according to the type of anode used. The 'Pre-baked' electrolytic cell uses multiple pre-baked carbon anodes that must be regularly replaced. The 'Søderberg' electrolytic cell uses a single continuous carbon anode, which is self-baked in situ within the cell by means of the heat released during the electrolytic process within the smelter; 'green' anode paste briquettes are added at the top while the anode is consumed at the bottom. Molten aluminium is deposited at the cathode and collects at the bottom of the cell, where it is periodically withdrawn by vacuum siphons into crucibles before being transported to the casting plant. At the casting plant molten aluminium is held in holding furnaces for further processing prior to casting metal ingots, blocks, billets, slabs or similar; small quantities of clean commercial scrap may also be added at this stage.

Annex I, point 3.17.2 ("Unwrought aluminium – primary aluminium from electrolytic reduction") to the Methodology Act defines the system boundaries for direct emissions monitoring for the primary (electrolytic) smelting production route, as encompassing:

> *" – All processes directly or indirectly linked to the production processes emitting CO_2 emissions from the consumption of electrodes or electrode pastes.*
>
> *– All processes directly or indirectly linked to the production processes emitting CO_2 emissions from any fuels used (e.g. for drying and pre-heating of raw materials, heating of electrolysis cells, heating required for casting).*
>
> *– All processes directly or indirectly linked to the production processes emitting CO_2 emissions from any flue gas treatment, from soda ash or limestone if relevant.*
>
> *– Perfluorocarbon emissions caused by anode effects monitored in accordance with Section B.7 of Annex II."*

There are no precursors for this production process. Indirect emissions resulting from electricity consumed by the production process are not taken into account in the embedded emissions of unwrought aluminium CBAM goods, in line with Article 7(1) and Annex II ("List of goods for which only direct emissions are to be taken into account") to the CBAM Regulation. Operators may nevertheless monitor electricity consumption and related indirect emissions, for example for internal purposes or under other monitoring and reporting schemes. In addition, in line with Annex I, point 3.1 (cross-sectoral rules), the system boundaries exclude the purchase and maintenance of infrastructure and equipment.

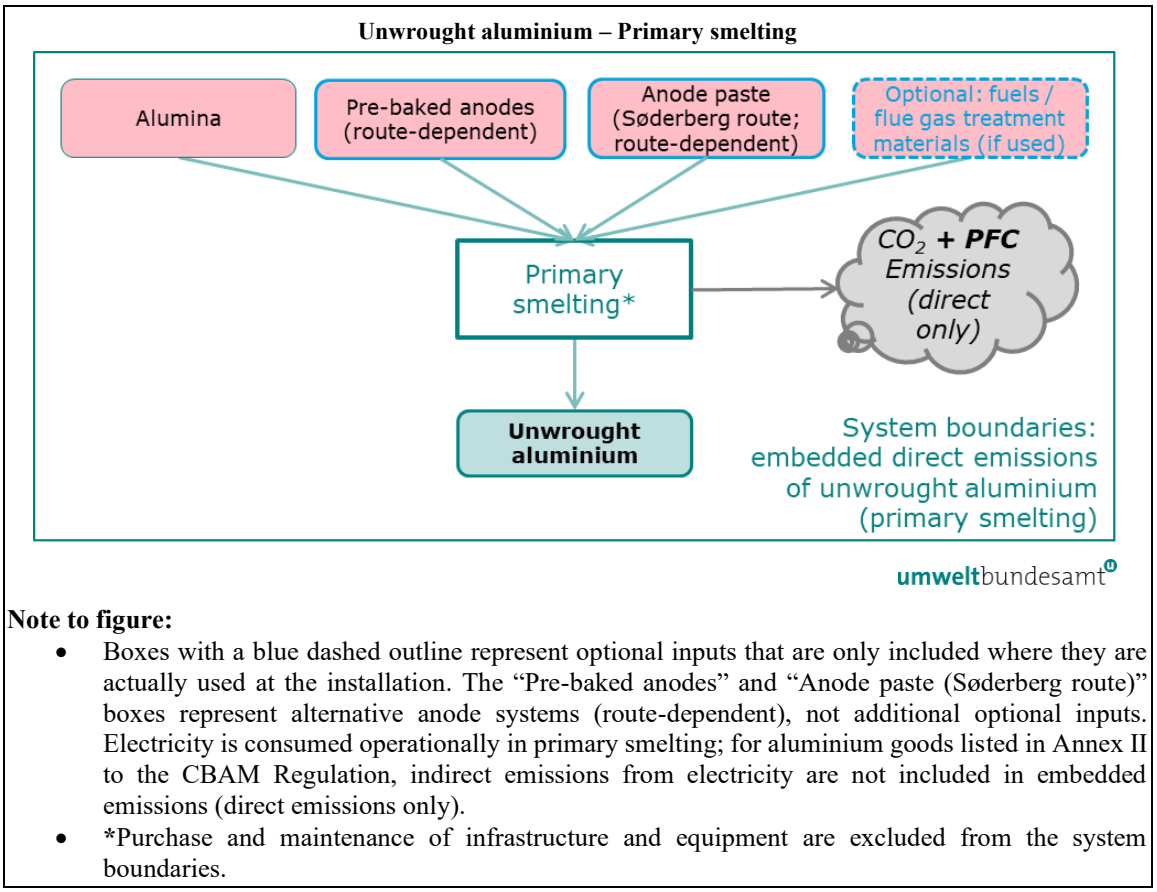
In line with the above definition of systems boundaries, the following production steps may be regarded as being within the system boundaries of primary aluminium installations:

- Raw material preparation – including storage of various additive constituents.

[6] Alumina is purified aluminium oxide produced by beneficiation of bauxite ore via the Bayer process. Production of alumina usually takes place at a different site to primary aluminium production for logistical and power supply reasons

- Electrolytic cell system for aluminium production process – all steps.
- Casting plant – all steps including holding furnaces, conveying systems, further metal processing (metal treatment, alloying and homogenisation) and casting.
- Emissions control – for treating releases to air, water or ground.
- The process materials consumed by the primary aluminium production route – alumina, pre-baked carbon anodes, 'green' anode paste briquettes, cryolite and other additives – are treated as raw materials and so have zero embedded emissions.

Figure 2-2: System boundaries of the unwrought aluminium – primary smelting production route



Note to figure:

- Boxes with a blue dashed outline represent optional inputs that are only included where they are actually used at the installation. The "Pre-baked anodes" and "Anode paste (Søderberg route)" boxes represent alternative anode systems (route-dependent), not additional optional inputs. Electricity is consumed operationally in primary smelting; for aluminium goods listed in Annex II to the CBAM Regulation, indirect emissions from electricity are not included in embedded emissions (direct emissions only).
- *Purchase and maintenance of infrastructure and equipment are excluded from the system boundaries.

2.2.3.2 Unwrought aluminium – Secondary melting (recycling) production route

Secondary aluminium is produced mainly from post-consumer aluminium scrap collected for recycling (although unwrought aluminium may also be separately added). Scrap is sorted according to type (cast or wrought alloy) and the sort of pre-treatment measures required (e.g. de-coating, de-oiling), and is then re-melted in the appropriate type of furnace (typically rotary or reverberatory, but induction furnaces may also be used) before further processing including: alloying, melt treatment (addition of salt or chlorination) and finally casting metal ingots, blocks, billets, slabs or similar. Typical fuels used are natural gas, LPG or fuel oil.

The Methodology Act (Annex I, point 3.17.3 ("Unwrought aluminium – secondary aluminium from melting) defines the system boundaries for direct emissions monitoring for the secondary melting (recycling) production route, as encompassing:

> *" – All processes directly or indirectly linked to the production processes emitting CO_2 emissions from any fuels used for drying and pre-heating of raw materials, used in melting furnaces, in pre-treatment of scrap such as de-coating and de-oiling, and combustion of the related residues, and fuels required for casting of ingots, billets or slabs.*
>
> *– All processes directly or indirectly linked to the production processes emitting CO_2 emissions from any fuels used in associated activities such as treatment of skimmings and slag recovery.*
>
> *– All processes directly or indirectly linked to the production processes emitting CO_2 emissions from any flue gas treatment, from soda ash or limestone if relevant."*

A precursor is unwrought aluminium from other sources, if used in the process. During the transitional period, indirect emissions resulting from electricity consumption were monitored and reported separately. In the definitive period, for aluminium CBAM goods listed in Annex II to the CBAM Regulation, indirect emissions from electricity are not included in embedded emissions (direct emissions only); operators may nevertheless monitor electricity consumption and related indirect emissions for internal purposes or under other monitoring and reporting schemes.

In line with the above definition of systems boundaries, the following production steps should be regarded as being within the system boundaries of secondary aluminium:

- Raw material preparation – including sorting, pre-treatment (de-coating, de-oiling), drying and pre-heating of scrap.
- Furnace system for aluminium production process – all steps, including furnace charging, melting and holding furnaces.
- Casting plant – all steps including holding furnaces, conveying systems, further metal processing (metal treatment, alloying and homogenisation) and casting.
- Emissions control – for treating releases to air, water or ground.

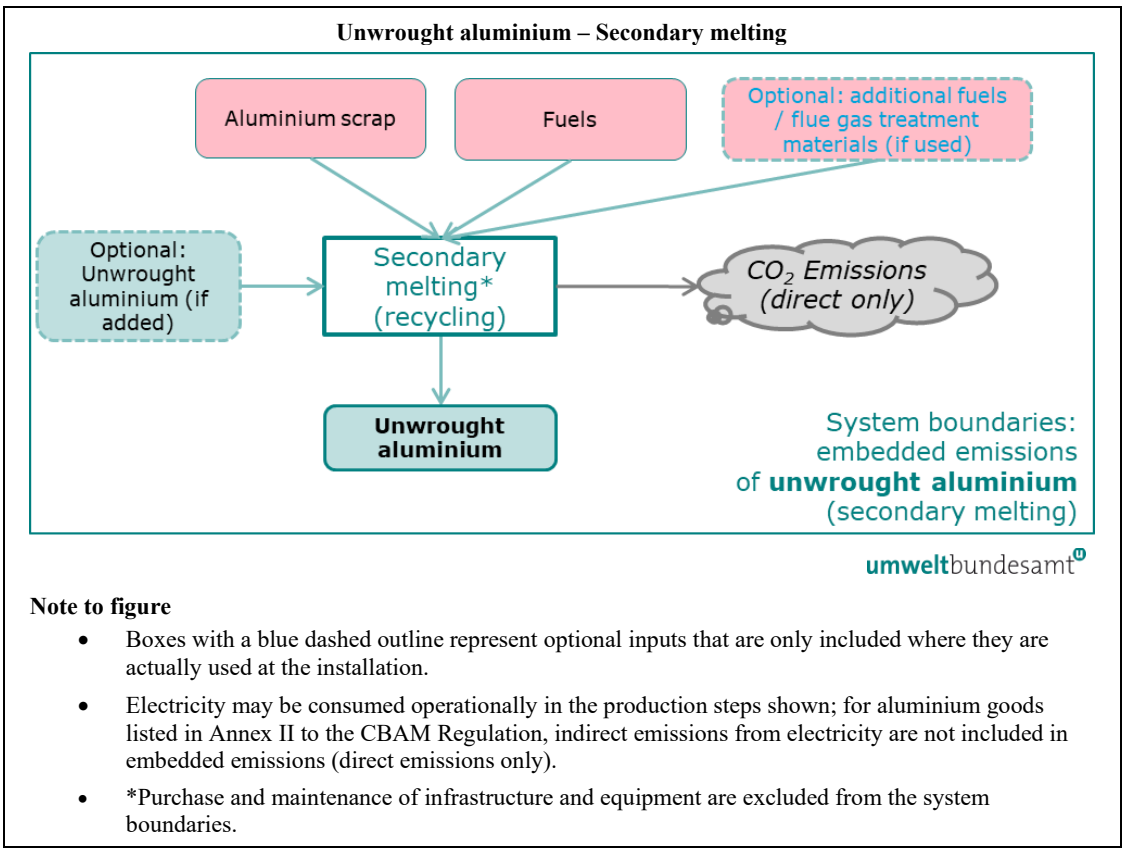
The following *Figure 2-3* shows the system boundaries of the relevant processes for secondary aluminium production.

There are no PFC emissions from the secondary aluminium process.

Aluminium scrap is the main material input to the secondary melting production route. Scrap (whether pre-consumer or post-consumer) is treated as a raw material and so has zero embedded emissions.

.

Figure 2-3: System boundaries of the Unwrought aluminium – secondary melting production route



Note to figure

- Boxes with a blue dashed outline represent optional inputs that are only included where they are actually used at the installation.
- Electricity may be consumed operationally in the production steps shown; for aluminium goods listed in Annex II to the CBAM Regulation, indirect emissions from electricity are not included in embedded emissions (direct emissions only).
- *Purchase and maintenance of infrastructure and equipment are excluded from the system boundaries.

2.2.3.3 Aluminium products production process

Aluminium products are produced by the further processing of precursor unwrought aluminium (alloyed or un-alloyed). Aluminium products are produced by a variety of forming processes including extrusion, casting, hot and cold rolling, forging and drawing. Extrusion is a common process used to produce aluminium profiles. Hot and cold rolling may be used to produce plate, sheet and foil. Casting may be used to produce complex forms.

The Methodology Act (Annex I, point 3.18 ("Aluminium products") defines the system boundaries for direct emissions monitoring for the aluminium products production route, as encompassing:

> *" – All processes directly or indirectly linked to the production processes emitting CO_2 emissions from combustion of fuels and process emissions from flue gas treatment, excluding the following processes: cutting, welding and finishing of aluminium products."*

Precursors are unwrought aluminium, if used in the production process (note that primary and secondary aluminium must be treated separately, if data is known, as each has different embedded emissions), and aluminium products, if used in the production process. Electricity consumption and related indirect emissions may be monitored (e.g. for internal purposes or under other monitoring and reporting schemes), but are not taken into account in embedded emissions for aluminium CBAM goods listed in Annex II to the CBAM Regulation (direct emissions only).

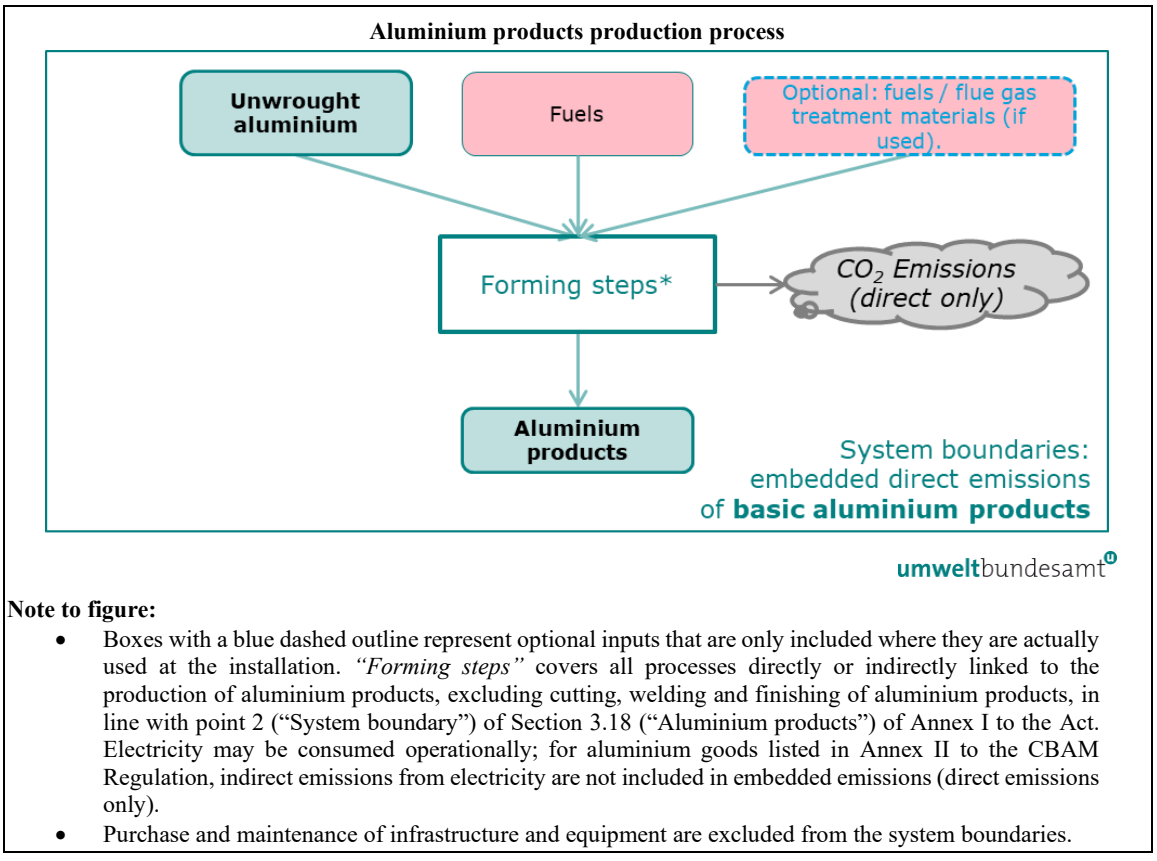
In line with the above definition of systems boundaries, the following production steps should be regarded as being within the system boundaries of basic aluminium products installations:

- Raw material preparation – including pre-heating, re-melting and alloying.
- Forming processes – all forming process steps for basic aluminium products, including (but not limited to): extrusion, casting, hot and cold rolling, forging, drawing.
- Annealing, surface preparation and treatment
- Emissions control – for treating releases to air, water or ground.

Purchase and maintenance of infrastructure and equipment are excluded from the system boundaries, in accordance with Annex I, point 3.1 (cross-sectoral rules) of the Methodology Act.

The following *Figure 2-4* shows the system boundaries of the relevant processes for aluminium products.

Figure 2-4: System boundaries of Aluminium products production process



Note to figure:

- Boxes with a blue dashed outline represent optional inputs that are only included where they are actually used at the installation. *"Forming steps"* covers all processes directly or indirectly linked to the production of aluminium products, excluding cutting, welding and finishing of aluminium products, in line with point 2 ("System boundary") of Section 3.18 ("Aluminium products") of Annex I to the Act. Electricity may be consumed operationally; for aluminium goods listed in Annex II to the CBAM Regulation, indirect emissions from electricity are not included in embedded emissions (direct emissions only).
- Purchase and maintenance of infrastructure and equipment are excluded from the system boundaries.

There are no PFC emissions resulting from aluminium products forming processes.

Also note that the quantity of aluminium products shall be reported in accordance with the applicable CN code and the functional unit (tonnes), without applying a standalone mass-threshold adjustment for non-aluminium components (e.g. lining or insulation).

3 ALUMINIUM SECTOR-SPECIFIC REQUIREMENTS

The textbox below signposts the sector-specific sections in the Methodology Act, relevant for the CBAM definitive period.

Methodology Act references:

- **Annex II, point 3** – Functional unit and system boundaries. Sub-sections 3.17 to 3.18 (aluminium sector aggregated goods categories).
- **Annex II, section G** – Global warming potentials for non-CO2 greenhouse gases.
- **Annex III, Section A** – Principles and general requirements, sub-section **A.4** Specific provisions of division of installations into production processes**.**
- **Annex III, Section B** – Monitoring of direct emissions at installation level, sub-section B.7. Requirements for determining perfluorocarbon emissions, covering: **B.7.1** Calculation method A – Slope Method; **B.7.2** Calculation method B – Overvoltage Method; **B.7.3** Determination of CO_2e emissions.

  Annex IV, point 2 – Sector-specific parameters to be included in the operator's emissions report.

3.1 Sector-specific requirements for monitoring and reporting

Direct embedded emissions should be monitored in line with the methodology set out in the Methodology Act.

3.1.1 Emissions monitoring

Relevant emissions that must be monitored and reported for the aluminium sector are:

- Carbon dioxide emissions (direct) resulting from the consumption of pre-baked carbon anodes or green anode paste during electrolysis – emissions result from the reaction of the carbon electrode with oxygen from the alumina or from other sources of oxygen such as from air[7]. There are also emissions associated with the self-baking (coking) of green anode paste in situ in the Søderberg process.

- Carbon dioxide emissions (direct) resulting from furnaces (e.g. holding, pre-heating, re-melting and annealing), where heated by the combustion of fuels used for the furnaces, from stationary plant only (excludes emissions from any mobile units such as vehicles).

- Carbon dioxide emissions (direct) resulting from the production of measurable heating (e.g. steam) and cooling, that is consumed within the system boundaries of the production process, regardless of the location of the production of the heating and cooling (i.e. from on-site generation or from imports from off-site).

[7] All formed carbon monoxide (CO) is assumed to be converted to CO_2.

- PFC (perfluorocarbon) emissions (direct) for CF_4 and C_2F_6 only, formed during brief upset conditions known as the 'Anode Effect', when alumina concentration levels drop too low and the electrolytic bath itself undergoes electrolysis.

- Carbon dioxide emissions (direct) resulting from emissions control (e.g. from carbonate raw materials such as soda ash used for acidic flue gas cleaning).

Note that emissions related to the production of pre-baked carbon anodes (even if produced on the same site) and alumina are excluded from the system boundary, i.e. they are not CBAM goods themselves and therefore no precursors with embedded emissions.

Direct emissions from the different source streams above are not reported separately but are added together to result in the total direct emissions for the installation or production process.

Installations that only carry out excluded processes for aluminium products

Where the installation only carries out processes excluded from the system boundaries (see section 3.18 of Annex I to the Methodology Act), verification obligations still apply if operators want to use actual values. Notably, the verification process will have to establish the following:

- That the installation only operates within the list of excluded processes
- Whether the production process carried out on the installation changed the specific embedded emissions of the goods:
    - for instance, in an installation where only cutting occurs, to compute embedded emissions of precursors, operators shall use the mass of precursors as **entering the production process** (before cutting), even if emissions from cutting activities will not be accounted for in embedded emissions of the complex good (see point B of Annex III to the Methodology Act). This means that SEE will increase, even if no emissions need to be reported for CBAM purposes.
- That embedded emissions of precursors, whether based on default values or actual values, are correctly calculated. For instance, the origin of precursors needs to be verified (to establish zero-rating if produced in the EU or exempted territories; to justify the use of the appropriate country-specific default values where relevant; to check that verification reports exist for precursors where relevant).

As the installation needs to be verified, the operator will have to establish a monitoring plan. In this case, the monitoring plan might be very concise given the absence of process emissions, combustion emissions, and indirect emissions. Emissions from excluded process steps do not have to be included in the monitoring plan as a source stream and do not have to be reported in the operator's emissions report.

3.1.2 Additional rules

Attribution of emissions

Given the complexity of production processes in the aluminium sector, installations producing two or more goods from the aggregated goods categories 'unwrought aluminium' or 'aluminium products', may be monitored and reported defining one joint production process for all the products from these groups covered, provided no intermediate product (i.e. precursor to the second processes) is sold or otherwise transferred out of the installation.

Simplified!

Determining PFC process emissions

Additional rules also apply for determining PFC emissions (CF_4 and C_2F_6 only) from primary aluminium production. However, where primary aluminium is used as a precursor, the related PFC emissions form part of the embedded emissions of the final product.

Two different calculation-based methods are available in accordance with Methodology Act, Annex II, section B.7. Both methods are considered equivalent, but as each requires different data, you should choose the method that is most appropriate for your installation's process control equipment:

- '**Slope method'** (**Method A**) – where the 'anode effect minutes per cell-day' (AEM) are recorded. The AEM expresses the frequency of anode effects (number of anode effects / cell-day) multiplied by the average duration of anode effects (anode effect minutes / occurrence).
- '**Overvoltage method'** (**Method B)** – where the 'anode effect overvoltage' (AEO) per cell [mV] is recorded. The AEO is determined as the integral of (time × voltage above the target voltage) divided by the time (duration) of data collection.

Calculation Method A – Slope Method

The following equations for determining PFC emissions shall be used under method A:

$$CF_4 \text{ emissions } [t] = AEM \times (SEF_{CF4}/1\,000) \times Pr_{Al} \quad \text{(Equation 21)}$$

$$C_2F_6 \text{ emissions } [t] = CF_4 \text{ emissions} \times F_{C2F6} \quad \text{(Equation 22)}$$

Where:

AEM is the anode effect minutes / cell-day;

SEF_{CF4} is the slope emission factor expressed in (kg CF_4 / t Al produced) / (anode effect minutes / cell-day)]. Where different cell-types are used, different *SEF* may be applied as appropriate;

Pr_{Al} is the production of primary aluminium [t] during the reporting period, and

F_{C2F6} is the weight fraction of C_2F_6 [t C_2F_6 / t CF_4].

The anode effect minutes per cell-day expresses the frequency of anode effects (number anode effects / cell-day) multiplied by the average duration of anode effects (anode effect minutes / occurrence):

$$AEM = frequency \times average\ duration \quad \text{(Equation 23)}$$

Emission factor: the emission factor for CF_4 (slope emission factor, SEF_{CF4}) expresses the amount [kg] of CF_4 emitted per tonne of aluminium produced per anode effect minute per cell-day. The emission factor (weight fraction F_{C2F6}) of C_2F_6 expresses the amount [kg] of C_2F_6 emitted proportionate to the amount [kg] of CF_4 emitted.

Table 3-1: Technology-specific emission factors related to activity data for the slope method.

Technology	**Emission factor for CF_4 (SEF_{CF4})** **[(kg CF_4/t Al) / (AE-min/cell-day)]**	**Emission factor for C_2F_6 (F_{C2F6})** **[t C_2F_6/ t CF_4]**
Legacy Point Feed Pre Bake (PFPB L)	0,122	0,097
Modern Point Feed Pre Bake (PFPB M)	0,104	0,057
Modern Point-Fed Prebake without fully automated anode effect intervention strategies for PFC emissions (PFPB MW)	– (*)	– (*)
Centre Worked Prebake (CWPB)	0,143	0,121
Side Worked Prebake (SWPB)	0,233	0,280
Vertical Stud Søderberg (VSS)	0,058	0,086
Horizontal Stud Søderberg (HSS)	0,165	0,077

(*) The installation has to determine the factor by own measurements. If this is technically not feasible or involves unreasonable costs, the values for CWPB methodology shall be used.

Calculation Method B – Overvoltage Method

For the overvoltage method, the following equations shall be used:

$$CF_4 \text{ emissions } [t] = OVC \times (AEO/CE) \times Pr_{Al} \times 0{,}001 \quad \text{(Equation 24)}$$

$$C_2F_6 \text{ emissions } [t] = CF_4 \text{ emissions} \times F_{C2F6} \quad \text{(Equation 25)}$$

Where:

OVC is the overvoltage coefficient ('emission factor') expressed in kg CF_4 per tonne of aluminium produced per mV overvoltage;

AEO is the anode effect overvoltage per cell [mV] determined as the integral of (time × voltage above the target voltage) divided by the time (duration) of data collection;

CE is the average current efficiency of aluminium production [%];

Pr_{Al} is the annual production of primary aluminium [t], and

F_{C2F6} is the weight fraction of C_2F_6 [t C_2F_6 / t CF_4].

The term *AEO/CE* (Anode effect overvoltage / current efficiency) expresses the time-integrated average anode effect overvoltage [mV overvoltage] per average current efficiency [%].

Table 3-2: Technology-specific emission factors related to overvoltage activity data.

Technology	**Emission factor for CF_4** **[(kg CF_4/t Al) / mV]**	**Emission factor for C_2F_6** **[t C_2F_6/ t CF_4]**
Centre Worked Prebake (CWPB)	1,16	0,121

Technology	Emission factor for CF_4 [(kg CF_4/t Al) / mV]	Emission factor for C_2F_6 [t C_2F_6/ t CF_4]
Side Worked Prebake (SWPB)	3,65	0,252

- **Minimum requirement** for both methods: technology-specific emission factors given in the Methodology Act, Annex II, section B.7 are used.
- **Recommended improvement** for both methods**:** Installation-specific emission factors for CF_4 and C_2F_6 are established through continuous or intermittent field measurements at least every 3 years or after significant changes[8] in the installation, taking into account industry best practice guidelines[9]. *The emission factors shall be determined with a maximum uncertainty of ±15% each.*

Calculating CO_2(e) emissions from PFC emissions

The following formula (Equation 26) may be used to calculate CO_2(e) from CF_4 and C_2F_6 emissions, using the global warming potential (GWP) for these gases:

$$PFC\ emissions\ [t\ CO_{2(e)}] = CF_4\ emissions\ [t] \times GWP_{CF4} + C_2F_6\ emissions\ [t] \times GWP_{C2F6}$$

Please see Table 6 in section G of Annex II to the Methodology Act for relevant GWP values as shown in *Table 3-3*.

Table 3-3: Global warming potentials

Gas	Global warming potential
N_2O	265 t CO_2e / t N_2O
CF_4	6 630 t CO_2e / t CF_4
C_2F_6	11 100 t CO_2e / t C_2F_6

Furthermore, fugitive emissions of PFC are taken into account calculated from the emissions measurable in a duct or stack ('point source emissions') using the collection efficiency of the duct:

$$PFC\ emissions\ (total) = PFC\ emissions\ (duct) / collection\ efficiency \quad \text{(Equation 20)}$$

The collection efficiency shall be measured when the installation-specific emission factors are determined.

[8] Relevant changes shall include a change in the distribution of anode effect duration, or a change in the control algorithm affecting the mix of the types of anode effects or the nature of the anode effect termination routine.

[9] In particular, the most recent best practice guidelines provided by the International Aluminium Institute.

3.1.3 Additional reporting requirements

The following Table 3-4 lists out the additional information that needs to be provided by you as an operator to importers, in your emissions data communication to them.

Table 3-4: Additional aluminium sector parameters requested in the CBAM declaration

Aggregated good category	**Reporting requirement in the annual report**
Unwrought aluminium	– Tonnes of scrap used for producing one tonne of the unwrought aluminium product. – % of scrap that is pre-consumer scrap. – Content of alloys in aluminium: If the total content of elements other than aluminium exceeds 1%, the total percentage of such elements.
Aluminium products	– Tonnes of scrap used for producing one tonne of the aluminium product. – % of scrap that is pre-consumer scrap. – Content of alloys in aluminium: If the total content of elements other than aluminium exceeds 1%, the total percentage of such elements.

These parameters depend on the goods produced. Alloying elements play a minor role and are not reflected in the CN classification of aluminium goods. Note that any alloying elements contained in unwrought aluminium, that are not defined as CBAM precursor goods, are excluded from the determination of mass and embedded emissions of the unwrought aluminium or resulting aluminium product good.

You need to ensure that you collect all the parameters necessary for your CBAM goods and communicate them to the importers of your goods. The importer will need to report the additional parameters when the goods are imported to the EU under the CBAM.

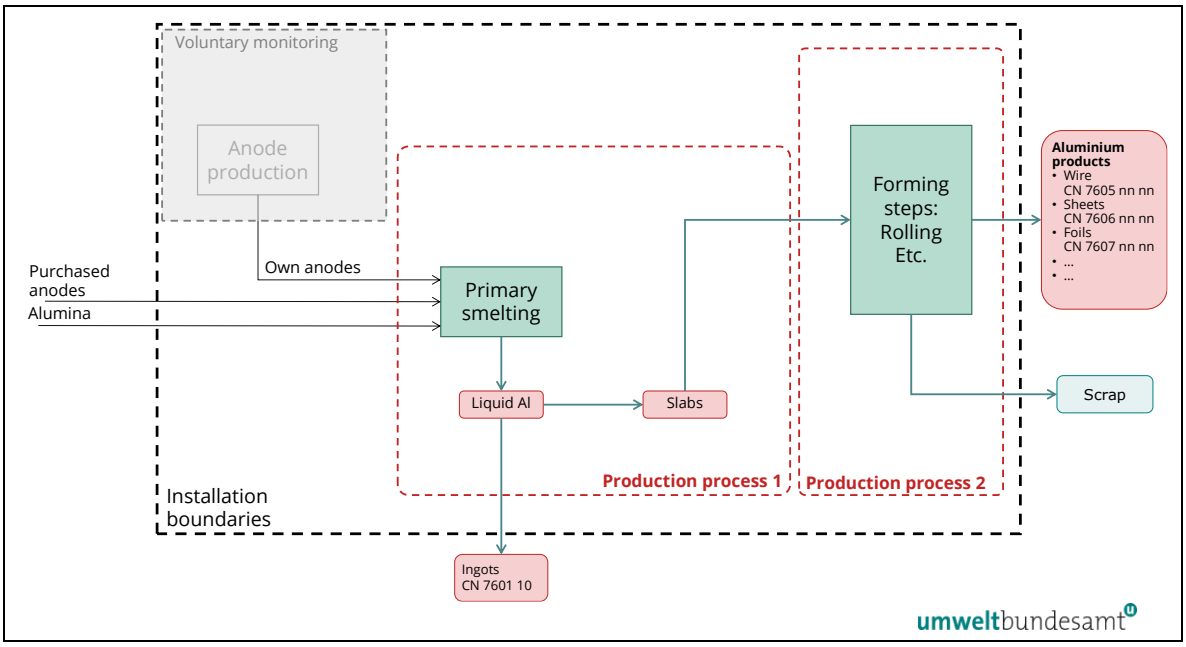
4 WORKED EXAMPLE FOR THE ALUMINIUM SECTOR

The following worked example shows how specific embedded emissions are derived for aluminium sector goods. The resulting embedded emissions of imports into the EU are then calculated at the end of the example for reporting in the definitive period. In this example the installation produces products from two aggregated goods categories, unwrought aluminium and aluminium products, each of which is defined as a single production process, as the intermediate product is sold. Therefore, a ‘bubble approach’ is not possible. All goods produced from the category “Aluminium products” belong to CN

codes that comply with the criterion to differ only in size or shape, but having the same composition. Therefore, these CN codes may be grouped into one production process, as in this example.

Figure 4-1 gives an outline view of the installation and shows the system boundaries as a hatched line for each production process. The physical units carrying out each production process have been grouped under 'Primary smelting' and 'Forming steps' and the different inputs and outputs and sources of emissions have been identified for each production process.

Figure 4-1: Aluminium example – overview



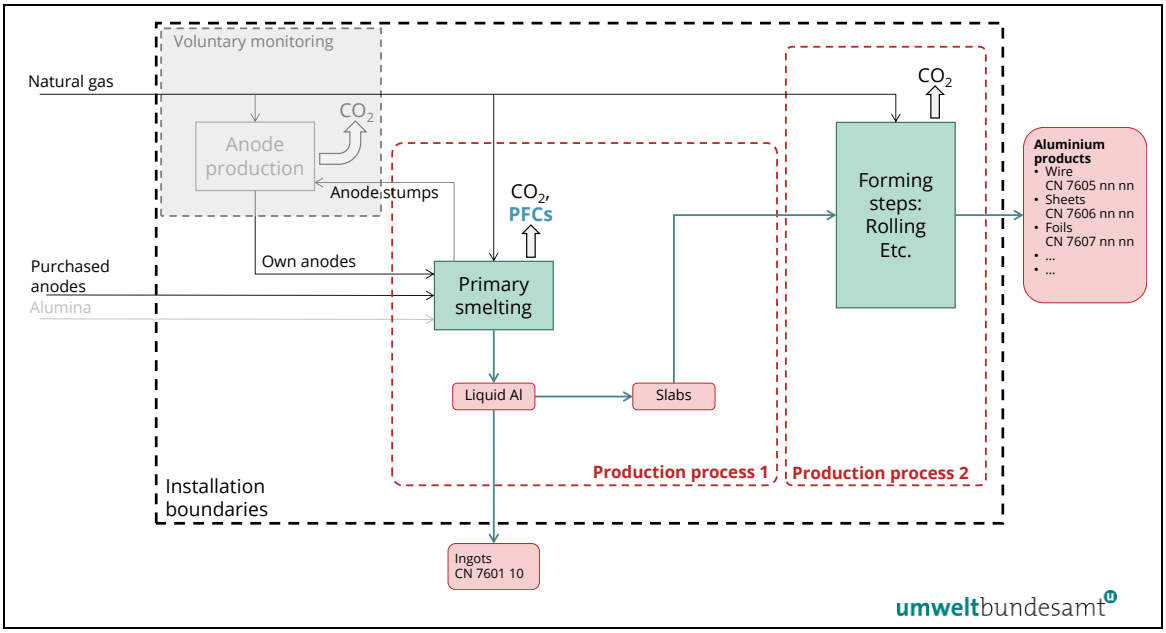
The two production processes defined above are:

- Production process 1 – primary smelting process route producing unwrought aluminium as ingots (which may be sold) and slabs, which are transferred to production process 2. Raw material inputs are anodes, both manufactured on site and purchased from elsewhere, and alumina.

- Production process 2 – different forming processes producing a range of aluminium products, such as wire, sheets and foils. Raw material inputs are unwrought aluminium slabs transferred from production process 1. There is also scrap from this process. This is sent off-site for recycling.

The second diagram (Figure 4-2) identifies sources of direct emissions from the installation.

Figure 4-2: Aluminium example – Identification of source streams for monitoring of direct emissions



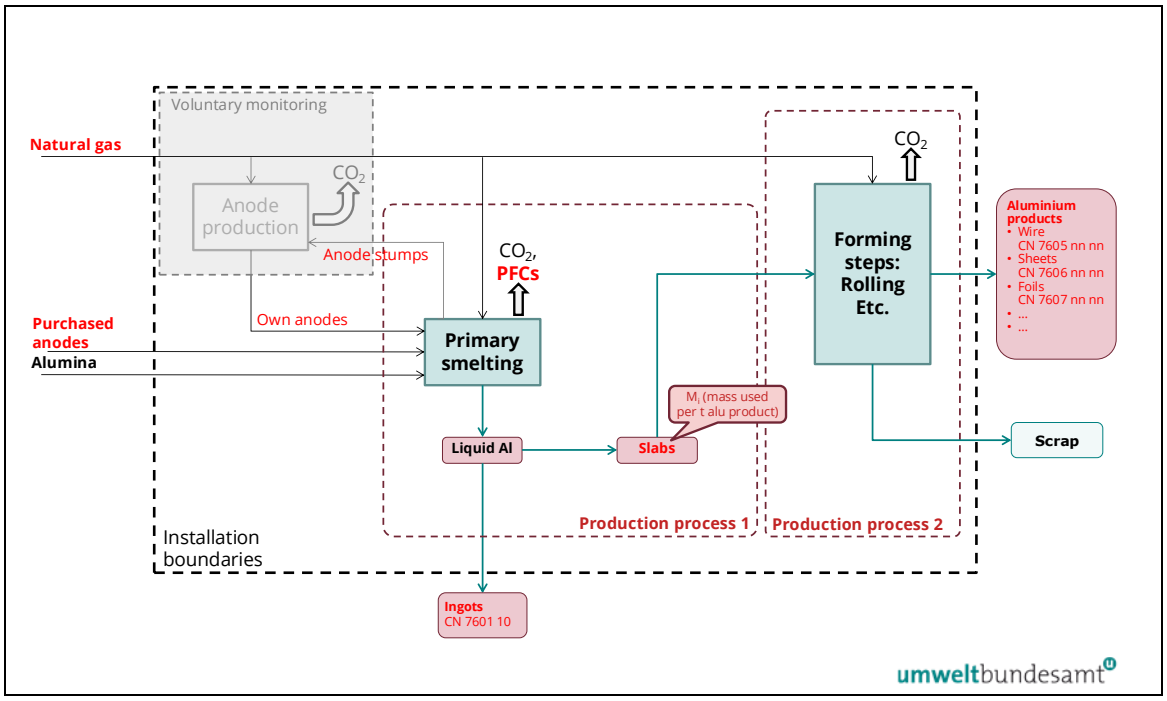
Direct emissions above result from fuel combustion in both production processes, and from the primary smelting process – from the consumption of carbon anodes and from the formation of PFCs.

Note that on-site anode production is omitted from monitoring, as anodes are raw materials and are therefore considered to have zero embedded emissions. For monitoring the anode consumption, the difference of anode input and anode stumps recycled result in the activity data of anode consumption.

However, for completeness you may wish on a voluntary basis to fully monitor all sources of direct and indirect emissions, which in this case would include a full mass balance of raw materials and additional fuels consumed in the anode production. Alumina consumption does not need monitoring, as it neither contributes to the direct emissions nor to embedded emissions.

The third diagram (Figure 4-3) provides a complete monitoring approach for all source streams, for the example installation.

Figure 4-3: Aluminium example – complete monitoring approach. All elements highlighted in red need to be monitored.



The inputs and outputs highlighted in red text in Figure 4-3 are the parameters that need to be monitored by the operator in order to attribute emissions and determine direct specific embedded emissions for both production processes.

The direct emissions that are monitored in this example result from:

- Direct CO_2 emissions from fuel (natural gas) combustion, and from the process resulting from the consumption of carbon anodes.
- Direct emissions of PFCs formed during the electrolysis process.
- In the case of production process 2, embedded emissions in precursors (the slabs produced in process 1).

The inputs of precursors (with embedded emissions) and the activity level of the aluminium goods produced by each production process also need to be monitored. There is no need to monitor the electricity consumption, as no indirect emissions are included for aluminium goods.

Table 4-1 summarises the inputs and outputs from the two production processes that are monitored in order to determine total direct specific embedded emissions.

Table 4-1: Input & production levels of the aluminium example

Production:	**Ingots & liquid aluminium, total**	**200 000 t**
	Ingots (sale):	80 000 t
	Primary aluminium into process 2 (slabs)	120 000 t
	Aluminium products (process 2)	
	Wire (CN 7605)	45 000 t
	Sheets (CN 7606)	60 000 t
	Foils (CN 7607)	8 000 t
	Total Aluminium products (process 2)	**113 000 t**
	Scrap[10] sold	7 000 t
Inputs:	Alumina	380 000 t
	Electrodes (sum self-produced and purchased, minus stumps)	69 000 t
	Natural gas (12 219 t for process 1, 1 962 t for process 2)	14 181 t

Whilst some unwrought aluminium is sold off site in the form of ingots (80 000 tonnes), 120 000 tonnes is used as a precursor in production process 2, and there is scrap of 7 000 tonnes at the end. No emissions are attributed to aluminium scrap, which therefore has zero embedded emissions.

Table 4-2 summarises the calculation of direct emissions and their attribution to each production process.

Table 4-2: Aluminium example – Installation's total direct emissions

Direct emissions CO_2e	**Emissions**	**Units**
From electrodes (using factor 3,664 t CO_2 / t C):	252 816	t CO_2
From natural gas (NCV = 48 GJ/t, EF=56,1 t CO_2 / TJ):	32 902	t CO_2
From PFCs (using a method described in section 3.1.2)	25 282	t CO_2e
Total Process 1 (primary aluminium)	**311 000**	**t CO_2e**
Total process 2 (final aluminium products), emissions from natural gas	5 283	t CO_2
Total direct emissions of the installation	**316 283**	**t CO_2**

Using the data in the tables above, direct specific embedded emissions are then calculated separately for each production process as shown in Table 4-3.

[10] Not a CBAM good

Table 4-3: Example calculation of specific embedded emissions of complex final aluminium goods

		Production levels (t)	Process total emissions (t CO_2e)	Mass ratio (M_i) of precursor (t / t)	SEE dir. (t CO_2e / t)
Process 1 (unwrought aluminium – ingots and slabs)					
	Product				
	Ingots	80 000			
	Slabs	120 000			
	Total	**200 000**	**311 000**		**1,555**
Process 2 (final aluminium products)					
Precursor	Slabs	120 000		1,062	1,651
Aluminium products		113 000	5 283		0,047
Total embedded emissions of final aluminium products					**1,698**

When calculating the total embedded emissions of final aluminium products above, the **mass ratio (M_i)** of the precursor is taken into. This is the mass of unwrought aluminium slabs consumed per tonne of aluminium products, and is calculated as:

- Mass slabs / mass aluminium products: 120 000 t / 113 000 t = 1,062 t / t (as above).

The direct SEE_i values of the precursor are then adjusted by this ratio i.e.:

- For SEE_i direct (precursor): 1,555 t CO_2 / t × 1,062 t / t = 1,651 t CO_2 / t.

Total direct specific embedded emissions of the final complex aluminium product are calculated by **adding** the *SEE* values of the precursor (adjusted by the M_i) to the emissions of the production process for aluminium products, as above.

Using the above approach, the CBAM reporting obligation due for the import of final aluminium product into the EU during the definitive period may then be determined; for example, for the import of 100 tonnes of basic aluminium product e.g. sheets:

- **Definitive period (report only):**
    - Direct embedded emissions = 100 t × 1,698 t CO_2 / t = 169,8 t CO_2

    Total: 169,8 t CO_2

Annex A List of definitions

Term	Definition
'Accuracy'	means the closeness of the agreement between the result of a measurement and the true value of the particular quantity or a reference value determined empirically using internationally accepted and traceable calibration materials and standard methods, taking into account both random and systematic factors
'Activity data'	means the amount of fuels or materials consumed or produced by a process relevant for the calculation-based methodology, expressed in terajoules (TJ), mass in tonnes or (for gases) volume in normal cubic metres, as appropriate
'Actual emissions'	means the emissions calculated based on primary data from the production processes of goods and from the production of electricity consumed during those processes as determined in accordance with the methods set out in Annex IV [to the Methodology Act]
'Activity level'	means the quantity of goods to which the same functional unit applies that are produced within the system boundaries of a production process during a reporting period
'Aggregated goods categories'	means aggregated goods categories pursuant to Table 1 of point 2 of Annex I. A group of CBAM goods with their CN codes, used for the purpose of defining system boundaries of production processes.
'Agricultural, aquaculture, fisheries and forestry residues'	means residues that are directly generated by agriculture, aquaculture, fisheries and forestry and that do not include residues from related industries or processing
'Authorised CBAM declarant'	means a person authorised by the competent authority in accordance with Article 17 [of the CBAM Regulation]. The 'authorised CBAM declarant' [which replaces the term 'reporting declarant' from the transitional period] may be importers of CBAM goods, or their indirect customs representatives, who have applied for and been granted the status of authorised CBAM declarant.
'Batch'	means an amount of fuel or material representatively sampled and characterised, and transferred as one shipment or continuously over a specific period of time
'Biomass'	means biomass as defined in Article 2, point (24) of the Renewable Energy Directive (RED II)[11]; it includes bioliquids and biofuels as defined in Article 2, point (32) and (33), biomass fuels as defined in Article 2, point (27) and biogas as defined in Article 2 point (28) of the RED II;

[11] Directive (EU) 2018/2001

Term	Definition
'Biomass fraction'	means the ratio of carbon stemming from biomass to the total carbon content of a fuel or material, expressed as a fraction
'Calculation factors'	means net calorific value, emission factor, preliminary emission factor, oxidation factor, conversion factor, carbon content or biomass fraction
'Calibration'	means the set of operations, which establishes, under specified conditions, the relations between values indicated by a measuring instrument or measuring system, or values represented by a material measure or a reference material and the corresponding values of a quantity realised by a reference standard
'Carbon price'	means the monetary amount paid in a third country, under a carbon emissions reduction scheme, in the form of a tax, levy or fee or in the form of emission allowances under a greenhouse gas emissions trading system, calculated on greenhouse gases covered by such a measure, and released during the production of goods
'CBAM certificate'	means a certificate in electronic format corresponding to one tonne of CO_2e of embedded emissions in goods
'CO_2 emission factor'	means the weighted average of the CO_2 intensity of electricity produced from fossil fuels within a geographic area. The CO_2 emission factor is the result of the division of the CO_2 emission data of the electricity sector by the gross electricity generation based on fossil fuels in the relevant geographic area. It is expressed in tonnes of CO_2 per megawatt-hour
'Combined nomenclature' (CN)	means the classification of goods, designed to meet the needs of: i) the Common customs tariff, setting import duties for products imported into the European Union (EU), as well as the Integrated tariff of the European Communities (Taric), incorporating all EU and trade measures applied to goods imported into and exported out of the EU; ii) the international trade statistics of the EU. The CN provides the means of collecting, exchanging and publishing data on EU international trade statistics. It is also used for the collection and publication of international trade statistics in intra-EU trade.[12]
'Combustion emissions'	means greenhouse gas emissions occurring during the exothermic reaction of a fuel with oxygen
'Competent authority'	means the authority designated by each Member State in accordance with Article 11 [of the CBAM Regulation]

[12] For definition see: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Combined_nomenclature_(CN)

Term	Definition
'Continuous emission measurement' (CEM)	means a set of operations having the objective of determining the value of a quantity by means of periodic measurements, applying either measurements in the stack or extractive procedures with a measuring instrument located close to the stack, whilst excluding measurement methodologies based on the collection of individual samples from the stack
'Control system'	means the operator's risk assessment and entire set of control activities, including the continuous management thereof, that an operator has established, documented, implemented and maintained pursuant to point A.2. of Annex II
'Complex goods'	means goods other than simple goods
'Conservative'	means that a set of assumptions is defined in order to ensure that no under-estimation of reported emissions or over-estimation of production of heat, electricity or goods occurs
'Conversion factor'	means the ratio of carbon emitted as CO_2 to the total carbon contained in the source stream before the emitting process takes place, expressed as a fraction, considering CO emitted to the atmosphere as the molar equivalent amount of CO_2
'Co-product'	means any of two or more products resulting from the same production process
'Customs declarant'	means a declarant as defined in Article 5, point (15) of Regulation (EU) No 952/2013 lodging a customs declaration for release for free circulation of goods in its own name or the person in whose name such a declaration is lodged
'CCUS system'	means a group of economic operators with technically connected installations and transport equipment for CO_2 capture, transport, use in production of goods, or geological storage
'Data flow activities'	mean activities related to the acquisition, processing and handling of data that are needed to draft an emissions report from primary source data
'Data set'	means one type of data, either at installation level or production process level as relevant in the circumstances, as any of the following: (a) the amount of fuels or materials consumed or produced by a production process as relevant for the calculation-based methodology, expressed in terajoules, mass in tonnes, or for gases as volume in normal cubic metres, as appropriate, including for waste gases; (b) a calculation factor;

Term	Definition
	(c) net quantity of measurable heat, and the relevant parameters required for determining this quantity, in particular: i) mass flow of heat transfer medium; and ii) enthalpy of transmitted and returned heat transfer medium, as specified by composition, temperature, pressure and saturation; (d) quantities of non-measurable heat, specified by the relevant quantities of fuels used for producing the heat, and the net calorific value (NCV) of the fuel mix; (e) quantities of electricity; (f) quantities of CO_2 transferred between installations; (g) quantities of precursors received from outside the production process, and their relevant parameters, such as country of origin, used production route, specific direct and indirect emissions;
'Default value'	means a value, which is calculated or drawn from secondary data, which represents the embedded emissions in goods
'Direct emissions'	means emissions from the production processes of goods including emissions from the production of heating and cooling that is consumed during the production processes, irrespective of the location of the production of the heating or cooling
'Embedded emissions'	means direct emissions released during the production of goods and indirect emissions from the production of electricity that is consumed during the production processes, calculated in accordance with the methods set out in Annex IV and further specified in the implementing acts adopted pursuant to Article 7(7)
'Emissions'	mean the release of greenhouse gases into the atmosphere from the production of goods
'Emission factor'	means the average emission rate of a greenhouse gas relative to the activity data of a source stream assuming complete oxidation for combustion and complete conversion for all other chemical reactions
'Emission factor' for electricity	means the default value, expressed in CO_2e, representing the emission intensity of electricity consumed in production of goods.
'Emission source'	means a separately identifiable part of an installation or a process within an installation, from which relevant greenhouse gases are emitted
EU ETS	means the system for greenhouse gas emissions allowance trading within the Union in respect of activities listed in Annex I to Directive 2003/87/EC other than aviation activities

Term	Definition
'EU ETS allowance'	means an allowance as defined in Article 3, point (a), of Directive 2003/87/EC in respect of activities listed in Annex I to that Directive other than aviation activities
'Fossil carbon'	means inorganic and organic carbon that is not biomass
'Fossil fraction'	means the ratio of fossil carbon to the total carbon content of a fuel or material, expressed as a fraction
'Fugitive emissions'	means irregular or unintended emissions from sources that are not localised, or too diverse or too small to be monitored individually
'Functional unit'	means the reference unit used for the calculation of embedded emissions in goods
'Goods'	means goods listed in Annex I [to the CBAM the CBAM Regulation]
'Greenhouse gases'	mean greenhouse gases as specified in Annex I [to the CBAM the CBAM Regulation] in relation to each of the goods listed in that Annex
'Heat meter'	means a thermal energy meter or any other device to measure and record the amount of thermal energy produced based upon flow volumes and temperatures
'Importer'	means either the person lodging a customs declaration for release for free circulation of goods or a bill of discharge in accordance with Article 175(5) of Delegated Regulation (EU) 2015/2446 in its own name and on its own behalf or, where the customs declaration is lodged by an indirect customs representative in accordance with Article 18 of Regulation (EU) No 952/2013, the person on whose behalf such a declaration is lodged
'Importation'	means release for free circulation as provided for in Article 201 of Regulation (EU) No 952/2013
'Indirect customs Representative'	A customs representative appointed by an importer established in a Member State. If the importer appoints an indirect customs representative and they agree, they can act as an authorised CBAM declarant and submit the application for an authorisation on behalf of the importer. The indirect customs representative is liable for all operations made for the importers it represents
'Indirect emissions'	means emissions from the production of electricity which is consumed during the production processes of goods, irrespective of the location of the production of the consumed electricity
'Inherent CO_2'	means CO_2 which is part of a source stream.
'Installation'	means a stationary technical unit where a production process is carried out

Term	Definition
'Legal metrological control'	means the control exercised by a public authority or regulator of the measurement tasks intended for the field of application of a measuring instrument, for reasons of public interest, public health, public safety, public order, protection of the environment, the levying of taxes and duties, the protection of consumers and fair trading
'Measurable heat'	means a net heat flow transported through identifiable pipelines or ducts using a heat transfer medium, such as, in particular, steam, hot air, water, oil, liquid metals and salts, for which a heat meter is or could be installed
'Measurement point'	means the emission source for which continuous emission measurement systems (CEMS) are used for emission measurement, or the cross-section of a pipeline system for which the CO_2 flow is determined using continuous measurement systems
'Measurement system'	means a complete set of measuring instruments and other equipment used to determine variables for the monitoring and calculation of emissions
'Minimum requirements'	means monitoring methods using the minimum efforts allowed for determining data in order to result in emission data acceptable for the purpose of the CBAM Regulation.
'Mixed fuel'	means a fuel which contains both biomass and fossil carbon
'Mixed material'	means a material which contains both biomass and fossil carbon
'Multifunctional process'	means a process that delivers multiple outputs or whose outputs feed into several production processes
'Net calorific value' (NCV)	means the specific amount of energy released as heat when a fuel or material undergoes complete combustion with oxygen under standard conditions, less the heat of vaporisation of any water formed
'Non-CBAM good'	means any good produced in the installation that is not included in Annex I to the CBAM Regulation
'Non-measurable heat'	means all heat other than measurable heat
'Operator'	means any person who operates or controls an installation in a third country, including a parent company that controls an installation in a third country
'Oxidation factor'	means the ratio of carbon oxidised to CO_2 as a consequence of combustion to the total carbon contained in the fuel, expressed as a fraction, considering carbon monoxide (CO) emitted to the atmosphere as the molar equivalent amount of carbon dioxide (CO_2)
'Power purchase agreement'	means a contract under which a person agrees to purchase electricity directly from an electricity producer

Term	Definition
'Precursor'	means any input material into a production process included in the list of goods set out in Annex I to the CBAM Regulation;
'Preliminary emission factor'	means the assumed total emission factor of a fuel or material based on the carbon content of its biomass fraction and its fossil fraction before multiplying it by the fossil fraction to produce the emission factor
'Production process'	means the chemical and physical processes carried out to produce goods in an installation
'Production route' [13]	means a specific technology used in a production process to produce goods
'Process emissions'	means greenhouse gas emissions other than combustion emissions occurring as a result of intentional and unintentional reactions between substances or their transformation, for a primary purpose other than the generation of heat, including from the following processes: (a) the chemical, electrolytic or pyrometallurgical reduction of metal compounds in ores, concentrates and secondary materials; (b) the removal of impurities from metals and metal compounds; (c) the decomposition of carbonates, including those used for flue gas cleaning; (d) chemical syntheses of products and intermediate products where the carbon bearing material participates in the reaction; (e) the use of carbon containing additives or raw materials; (f) the chemical or electrolytic reduction of metalloid oxides or non-metal oxides such as silicon oxides and phosphates.
'Proxy data'	means annual values which are empirically substantiated or derived from accepted sources and which an operator uses to substitute a data set[14] for the purpose of ensuring complete reporting
'Rebate'	means any amount that reduces the amount due or paid by a person liable for the payment of a carbon price, before its payment or after, in a monetary form or in any other form.
'Recommended improvements'	means monitoring methods which are proven means to ensure that data are more accurate or less prone to mistakes than by mere application of minimum requirements
'Related emissions'	means: – the direct emissions caused by the source stream or emission source concerned; – emissions attributed to a quantity of measurable heat; – the indirect emissions related to the quantity of electricity concerned; – embedded emissions of a material produced or of a precursor consumed.

[13] Note that different production routes can fall within the same production process.

[14] Refers to the activity data or the calculation factors.

Term	Definition
'Reporting period'	means the period corresponding to the calendar year during which the good was produced and used by the authorised CBAM declarant as reference for the determination of embedded emissions
'Residue'	means a substance that is not the end product that a production process directly seeks to produce; it is not a primary aim of the production process and the process has not been deliberately modified to produce it
'Simple goods'	means goods produced in a production process requiring exclusively input materials (precursors) and fuels having zero embedded emissions
'Source stream'	means either of the following: (a) a specific fuel type, raw material or product giving rise to emissions of relevant greenhouse gases at one or more emission sources as a result of its consumption or production; (b) a specific fuel type, raw material or product containing carbon and included in the calculation of greenhouse gas emissions using a mass-balance method
'Specific embedded emissions'	means the embedded emissions of one tonne of goods, expressed as tonnes of CO_2e emissions per tonne of goods
'Standard conditions'	means temperature of 273,15 K and pressure conditions of 101 325 Pa defining normal cubic metres (Nm^3)
'System boundary'	means the group of chemical or physical processes included in the calculation of embedded emissions of goods under the same aggregated goods category. The system boundaries shall cover direct emissions, indirect emissions for goods not listed in Annex II to the CBAM Regulation, and the embedded emissions of any precursor.
'Third country'	means a country or territory outside the customs territory of the [European] Union
'Tonne of $CO_2(e)$'	means one metric tonne of carbon dioxide ('CO_2'), or an amount of any other greenhouse gas listed in Annex I [to the CBAM Regulation] with an equivalent global warming potential ['CO_2e']
'Transmission system operator'	means an operator as defined in Article 2, point (35) of Directive (EU) 2019/944 of the European Parliament and of the Council ([15]).

[15] Directive (EU) 2019/944 of the European Parliament and of the Council of 5 June 2019 on common rules for the internal market for electricity and amending Directive 2012/27/EU (OJ L 158, 14.6.2019, p. 125).

Term	Definition
'Uncertainty'	means a parameter, associated with the result of the determination of a quantity, that characterises the dispersion of the values that could reasonably be attributed to the particular quantity, including the effects of systematic as well as of random factors, expressed in percent, and describes a confidence interval around the mean value comprising 95% of inferred values taking into account any asymmetry of the distribution of values
'Waste'	means any substance or object which the holder discards or intends or is required to discard, excluding substances that have been intentionally modified or contaminated in order to meet this definition
'Waste gas'	means a gas containing incompletely oxidised carbon in a gaseous state under standard conditions which is a result of any of the processes listed under 'process emissions'

Annex B List of abbreviations

Abbreviation	Full term
AD	Activity Data
AEM	Anode Effect Minutes
AEO	Anode Effect Overvoltage
AL	Activity Level
AOD	Argon Oxygen Decarburisation.
BAT	Best Available Techniques
BF	Biomass Fraction
BFG	Blast Furnace Gas
BM	Benchmark
BOF	Basic Oxygen Furnace
BOFG	Basic Oxygen Furnace Gas
BREFs	Best Available Techniques Reference documents
CA	Competent Authority
CBAM	Carbon Border Adjustment Mechanism, the
CCR	Clinker to Cement Ratio
CCS	Carbon Capture and Storage
CCU(S)	Carbon Capture and Utilisation (and Storage)
CEMS	Continuous Emission Measurement Systems
CF	Conversion Factor
CFP	Carbon footprint of products
CHP	Combined Heat and Power
CKD	Cement Kiln Dust
CN	Combined nomenclature
COG	Coke Oven Gas
DRI	Direct Reduced Iron
EAF	Electric Arc Furnace
EEA	European Economic Area (EU-27, Iceland, Norway, Liechtenstein)
EF	Emission Factor
EFTA	European Free Trade Area
EORI	Economic Operator Registration and Identification
ETS	Emissions Trading System
EU ETS	EU Emissions Trading System
EUA	EU Allowances (used in the EU ETS)
EUR	Euro (currency)

Abbreviation	Full term
FAR	Free Allocation Rules (Regulation 2019/331)[16]
FIFO	First-In-First-Out principle
GHG	Greenhouse Gas
GWP	Global Warming Potential
HBI	Hot Briquetted Iron
HS	Harmonised System (for international trade)
IEA	International Energy Agency
ISO	International Organization for Standardization
LULUCF	Land-use, land change and forestry (criteria)
MP	Monitoring Plan
MRR	Monitoring and Reporting Regulation (Regulation 2018/2066)[17]
MRV	Monitoring, Reporting and Verification
MS	Member State(s)
MWh	Megawatt-hour
NCV	Net Calorific Value
NPI	Nickel pig iron
OF	Oxidation Factor
PCI	Pulverised Coal Injection
PEMS	Predictive Emission Monitoring System
PFC	Perfluoro-carbon
PoS	Proofs of Sustainability
RED II	Renewable Energy Directive, re-cast
SEE	Specific embedded emissions
SEFA	Specific embedded free allocation
TARIC	Integrated Tariff of the European Union database
TJ	Terajoules
TSO	Transmission System Operator
UCC	Union Custom Code
UN/LOCODE	United Nations Code for Trade and Transport Location

[16] Free Allocation Rules (Commission Delegated Regulation (EU) 2019/331 of 19 December 2018 determining transitional Union-wide rules for harmonised free allocation of emission allowances pursuant to Article 10a of Directive 2003/87/EC of the European Parliament and of the Council)

[17] Monitoring and Reporting Regulation (Commission Implementing Regulation (EU) 2018/2066 of 19 December 2018 on the monitoring and reporting of greenhouse gas emissions pursuant to Directive 2003/87/EC of the European Parliament and of the Council and amending Commission Regulation (EU) No 601/2012